



# Memphis Zoo Custodial System

Operational design, screen walkthrough, and cost comparison

A practical view of how the custodial system connects daily area ownership, scan-based work confirmation, guest issue reporting, events, reminders, messaging, and Memphis AI into one working operations flow.

## 01 See the work

Managers can see scan activity, completion status, guest issues, and schedules from the screens they already use.

## 02 Guide the day

Custodians get simple phone workflows: assigned areas, scan prompts, event notices, due soon reminders, and issue notifications.

## 03 Keep proof simple

NFC/QR activity creates a time-stamped trail without asking employees to manage spreadsheets or paperwork.

## 04 Control cost

The design uses low-cost phones, NFC tags, kiosk licenses, GitHub Pages, Supabase, Render, and zoo Wi-Fi instead of broad commercial software licensing.

### Important framing

The system does not replace manager inspections. It tells managers when and where cleaned areas are ready to inspect, so checks can happen before guest traffic re-soils the area.

### Document style

This version uses the normal print/PC page layout and full application screenshots. It is not a phone-friendly image deck.

# How the System Works

The core idea is simple: a scan or report becomes a visible operations signal, and the right person gets notified.

## 1. Area ownership

Schedules assign custodians to restroom, exhibit, and service areas for the day.

## 2. Work starts at the location

The custodian uses the phone workflow and NFC/QR scan points to identify the real location.

## 3. Local-first activity

If Wi-Fi is weak, the phone keeps working locally and queues the action.

## 4. Sync and visibility

When connection returns, queued activity syncs to the backend and dashboard views refresh.

## 5. Messenger notifications

Guest issues, event notices, due soon locations, and overdue locations create Memphis Messenger alerts.

## 6. Manager inspection

Managers use the screens to know what is ready to inspect; the software supports the inspection, it does not replace it.

The system is strongest because the pieces are connected. Commercial tools often handle inspections, work orders, QR requests, or schedules separately. This design ties the custodial workflow together around the actual zoo day: staffing, locations, guests, events, scans, messages, and manager follow-up.

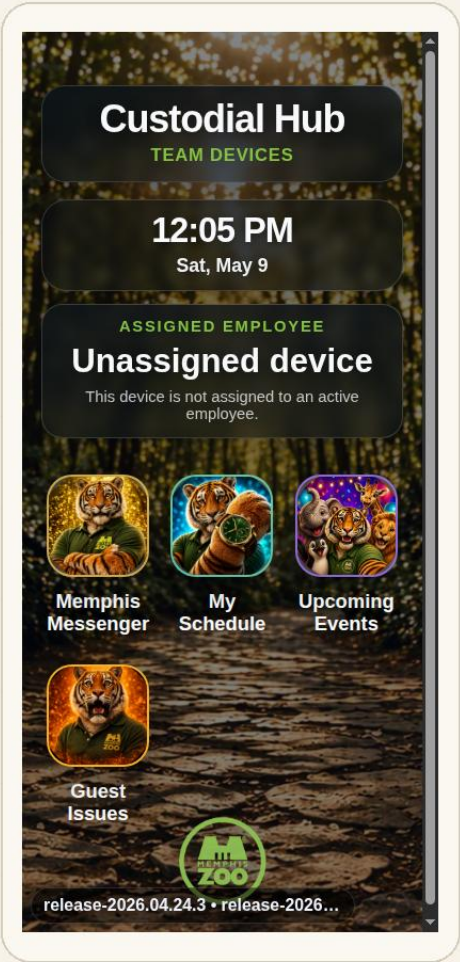
## WHAT CHANGES OPERATIONALLY

- Less guessing about which areas are ready for inspection.
- Less dependence on radio calls and memory when mornings get busy.
- Clearer response path when a guest reports a cleanliness issue.
- Event information reaches the people assigned to the affected area.
- Due soon and overdue reminders help prevent quiet misses.

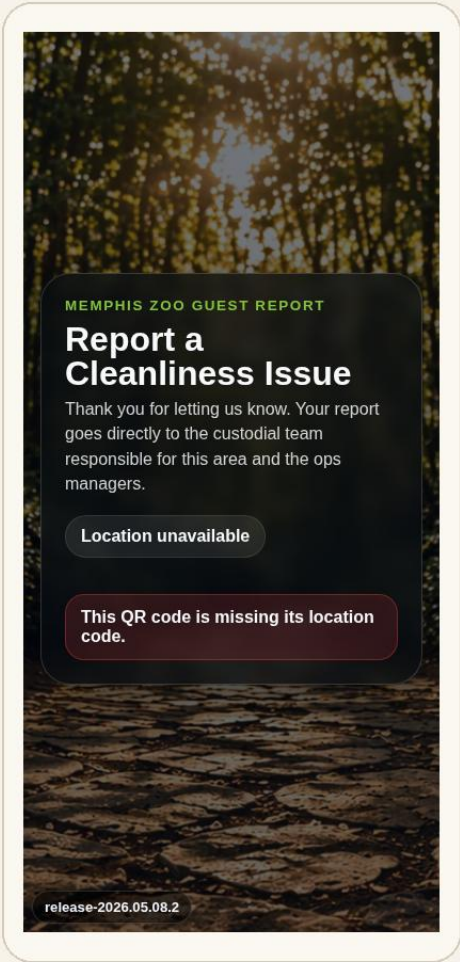


# Custodian Phone Views

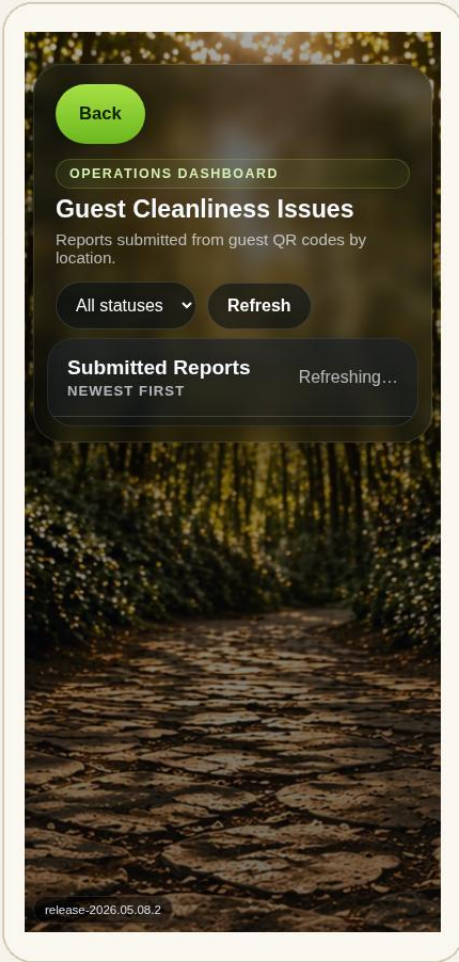
The phone side is intentionally simple: open the hub, choose the task, scan or report, and keep moving.



Employee Hub



Guest QR Report



Guest Issues View

## HOW THE PHONE WORKFLOW SHOULD FEEL

- Custodians do not need to understand the backend. They see clear tiles, schedules, messages, events, and guest issues.
- A guest QR issue becomes a Memphis Messenger notification so the assigned custodian and operations contact can respond.
- The same notification pattern supports events in the employee’s area, due soon reminders, and overdue location reminders.
- The workflow is built for zoo Wi-Fi. If connection drops, work can continue locally and sync when service returns.

### Design target

The phone screens are for quick action in the field. The PC screens are for planning, visibility, and follow-up.

PC view for daily visibility: cleaned areas, scan activity, open issues, and inspection readiness.

Back



OPERATIONS DASHBOARD

Custodial Tracking System

TODAY  
12:31 PM  
Sat, May 9, 2026  
CURRENT ATTENDANCE  
4,349

Restrooms Overdue  
0

Restrooms Not Cleaned  
30

Open Tickets  
4

Exhibits Overdue  
0

Exhibits Not Cleaned  
15

Restroom Status Report

LIVE OPERATIONAL STATE

| LOCATION                             | STATUS      | TIME | TIME TO CLEAN | SERVICES PERFORMED | EMPLOYEE NAME |
|--------------------------------------|-------------|------|---------------|--------------------|---------------|
| Cathouse Cafe Men's Restroom         | NOT CLEANED | —    | —             | —                  | —             |
| Expo Men's Restroom                  | NOT CLEANED | —    | —             | —                  | —             |
| West Admin Upstairs Women's Restroom | NOT CLEANED | —    | —             | —                  | —             |
| Bonobos Men's Restroom               | NOT CLEANED | —    | —             | —                  | —             |
| Bonobos Women's Restroom             | NOT CLEANED | —    | —             | —                  | —             |
| Primate House Men's Restroom         | —           | —    | —             | —                  | —             |

Exhibit Status Report

LIVE OPERATIONAL STATE

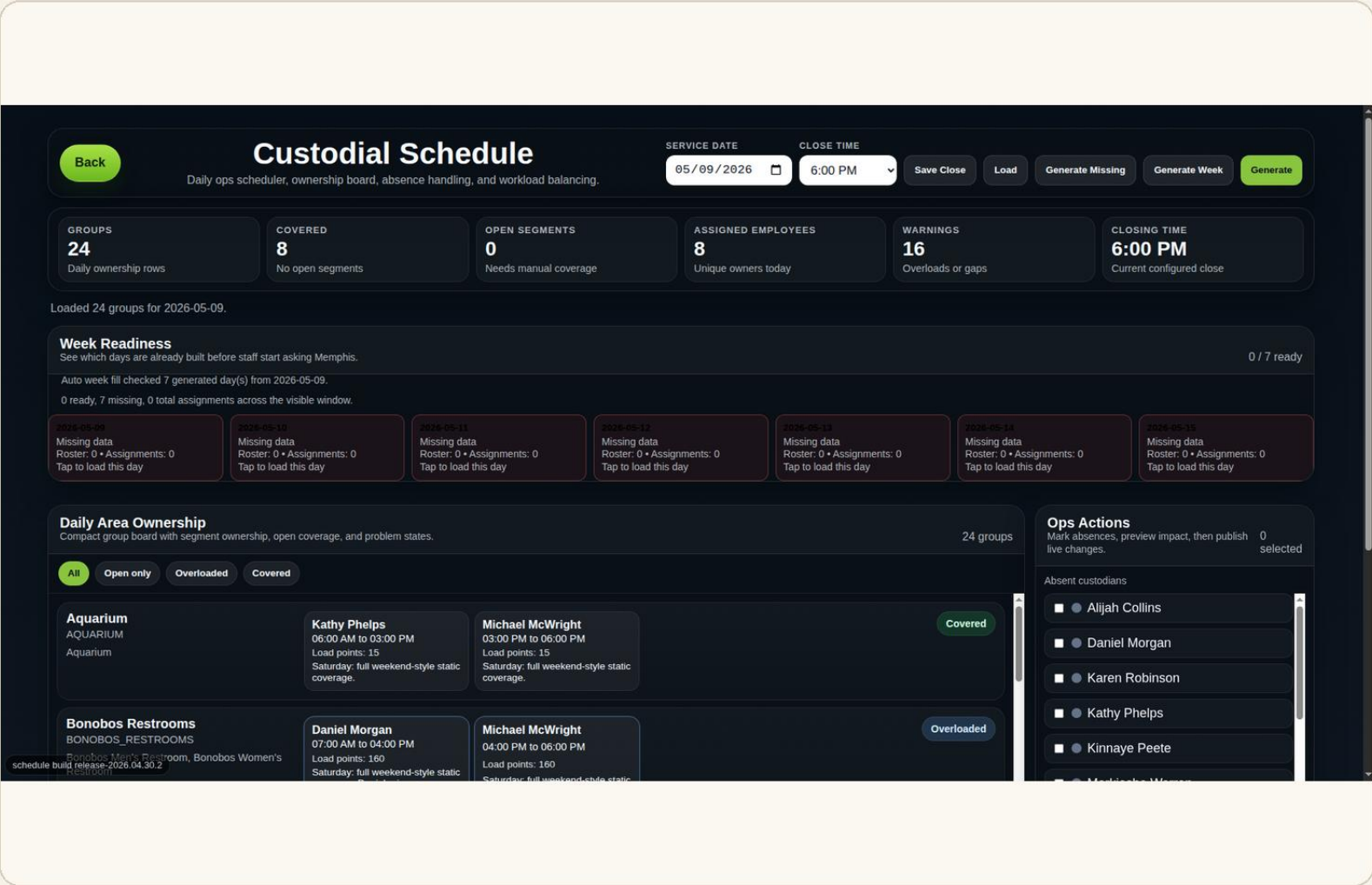
WHAT THIS SCREEN SHOWS

- Shows the manager-facing control view rather than asking someone to track restroom/exhibit status manually.
- Refresh behavior keeps the screen current while the day is moving. The screenshot is used as a screen example, not a claim that rollout is complete everywhere.
- The goal is not to replace inspection; it helps managers decide where inspection attention should go next.
- The value is quick visibility: what is active, what needs follow-up, and what is ready for a manager to check.



Daily Schedule and Absence Reassignment05

PC view for area ownership, close time, absence handling, and coverage planning.

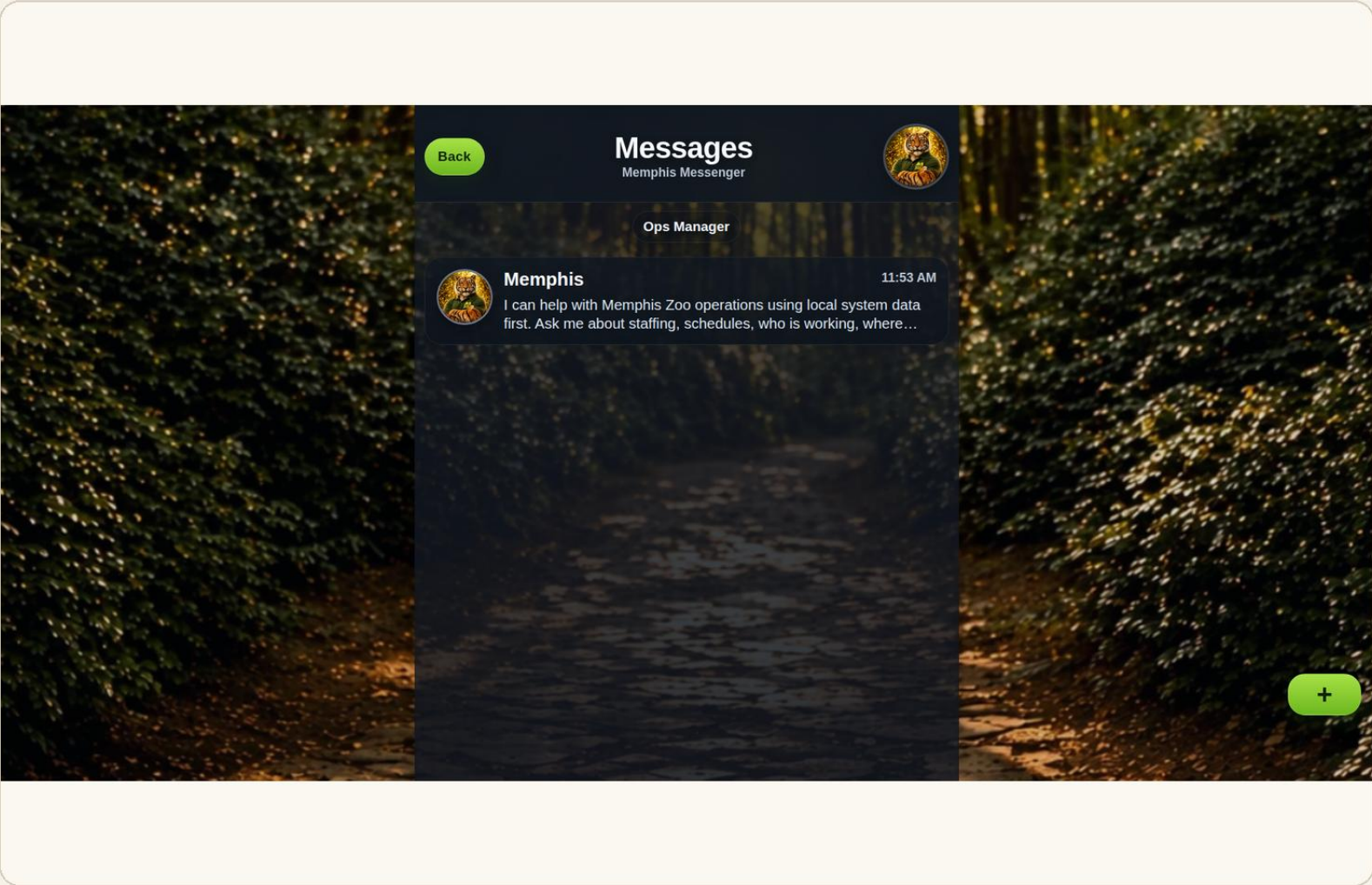


WHAT THIS SCREEN SHOWS

- Daily area ownership connects employees to location groups before the day begins.
- Absence tools preview and apply reassignment instead of leaving coverage decisions buried in notes or memory.
- Close-time and workload settings give the system context for reminders and coverage expectations.
- This keeps coverage decisions tied to the same system that handles scan activity and reminders.

# Memphis Messenger

PC view for internal messages and group conversations.



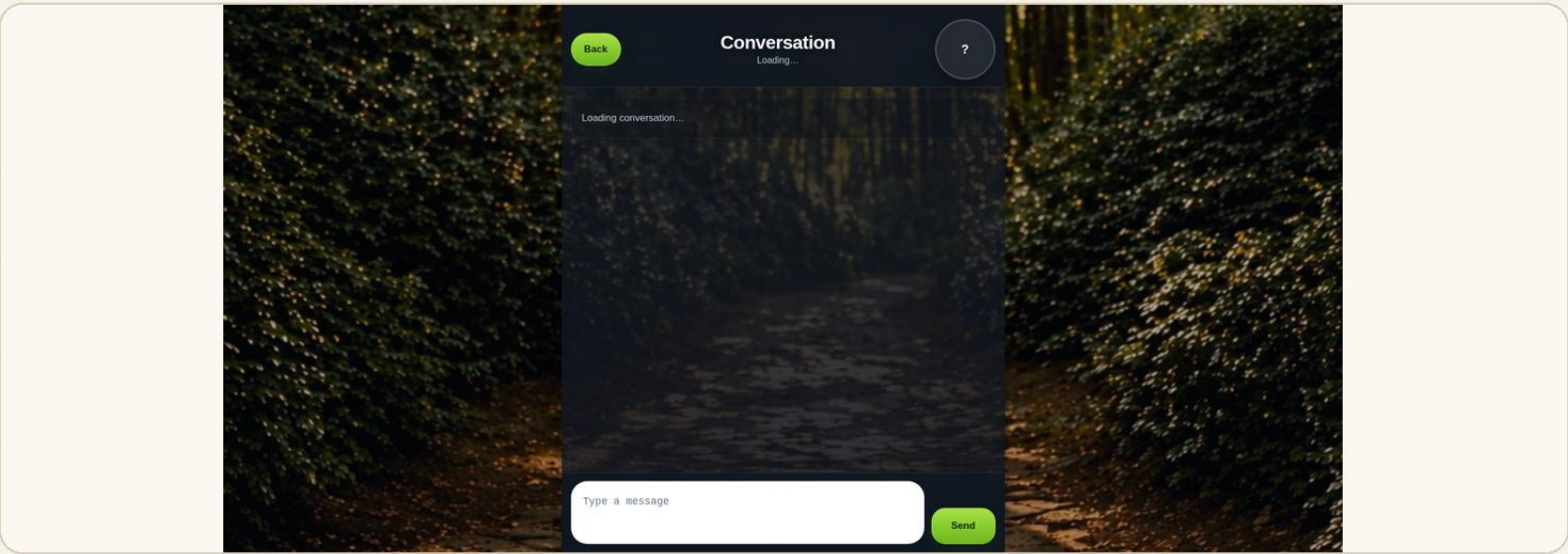
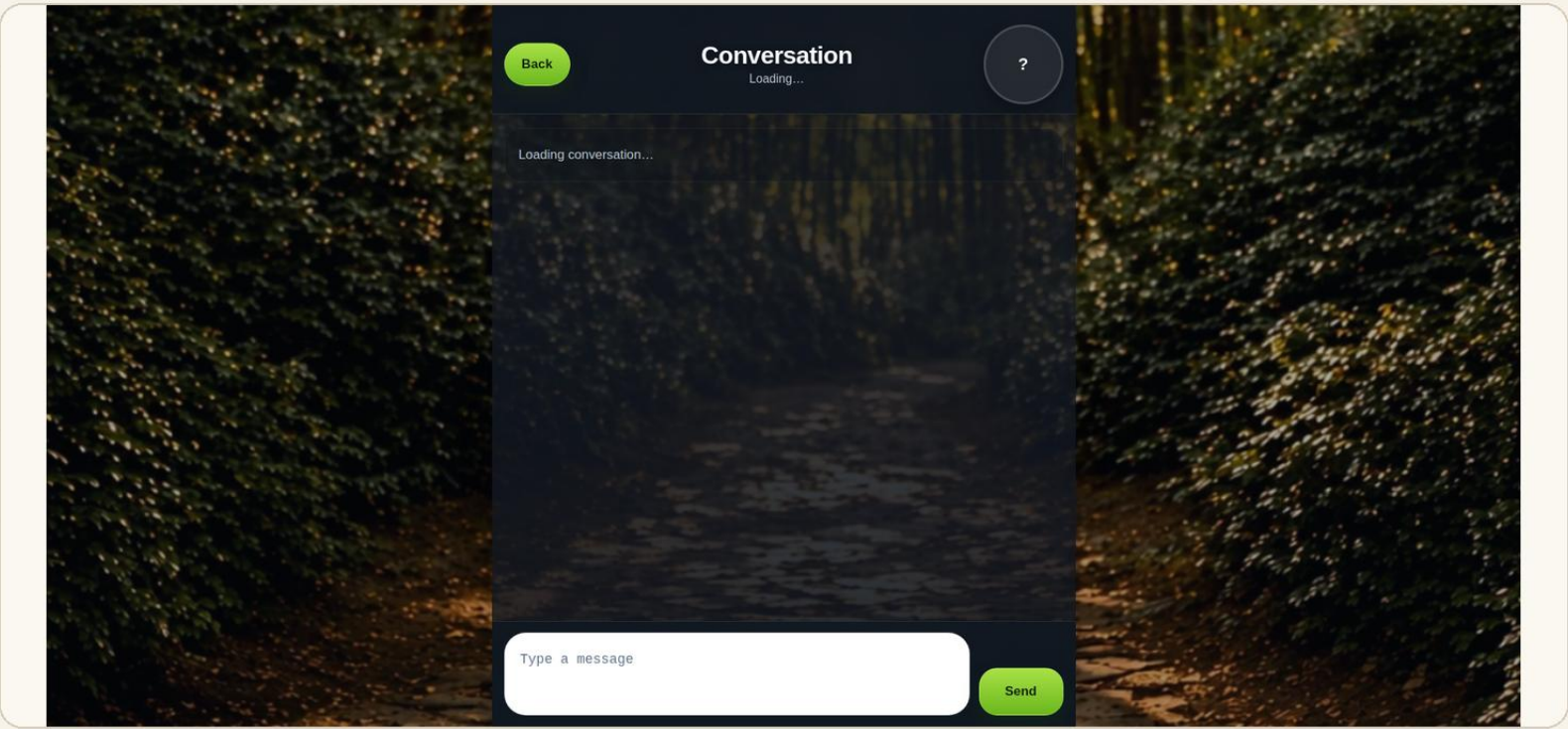
## WHAT THIS SCREEN SHOWS

- Messenger is the notification path for operational prompts, not just general chat.
- Guest issues can trigger Memphis Messenger notifications to the assigned custodian and operations contact.
- Events in an assigned area, due soon locations, and overdue locations follow the same notification concept.
- This gives staff a direct place to receive prompts and ask follow-up questions during the day.



# Memphis AI and Messenger

Messaging is not separate from operations; it is how the system turns data into timely reminders and answers.

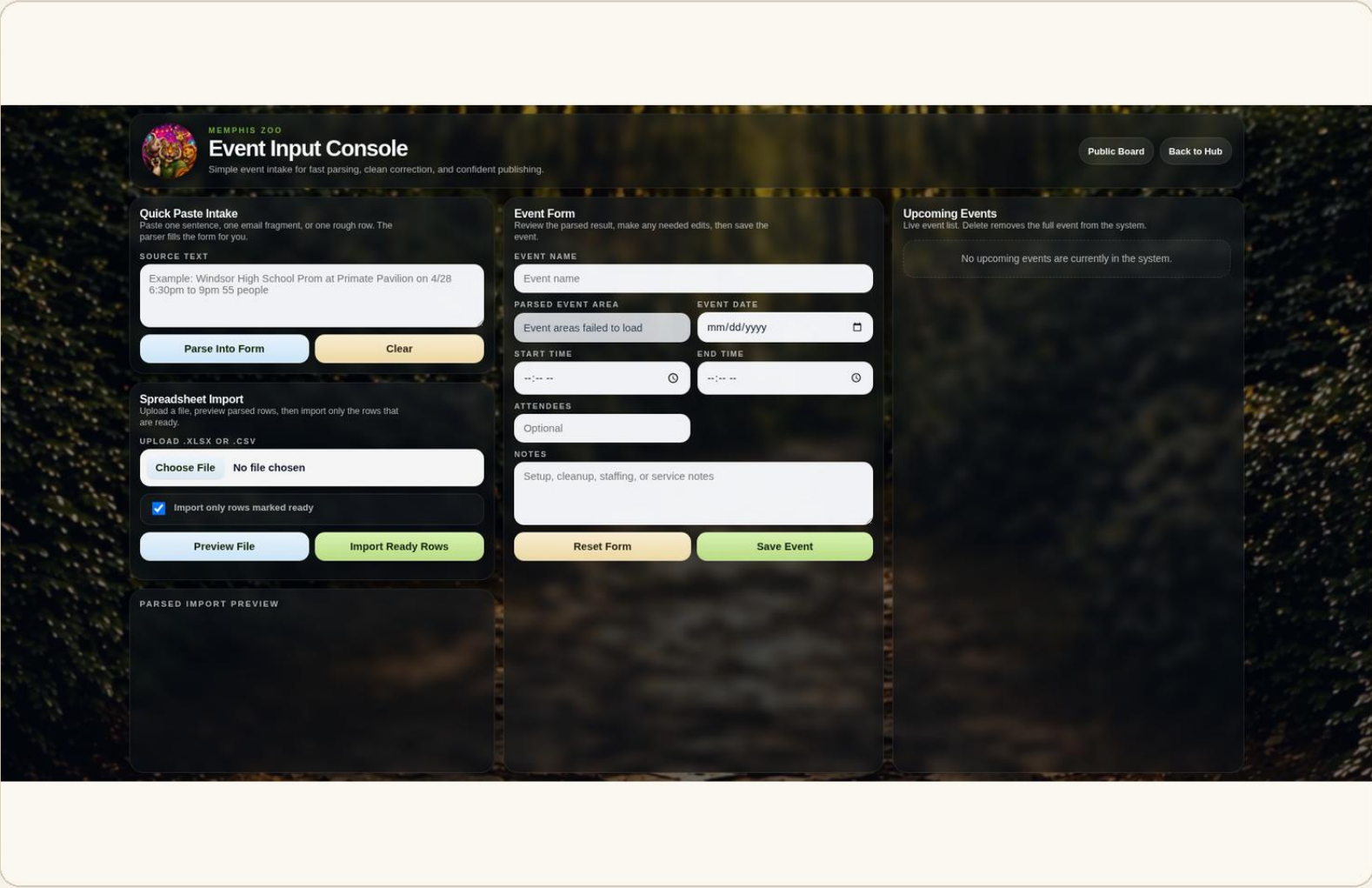


## WHAT MEMPHIS ADDS

- Answers operational questions from system data: staffing, schedules, area coverage, tickets, scans, events, attendance, and contacts.
- Keeps routine answers inside Memphis Messenger instead of forcing a manager to hunt through several screens.
- Supports notification workflows: guest issue reported, event in assigned area, due soon location, and overdue location.
- The live system can answer based on real assignments and current data, which makes it useful during the actual workday.

# Event Input Console

PC view for entering events by paste, spreadsheet import, or manual form.



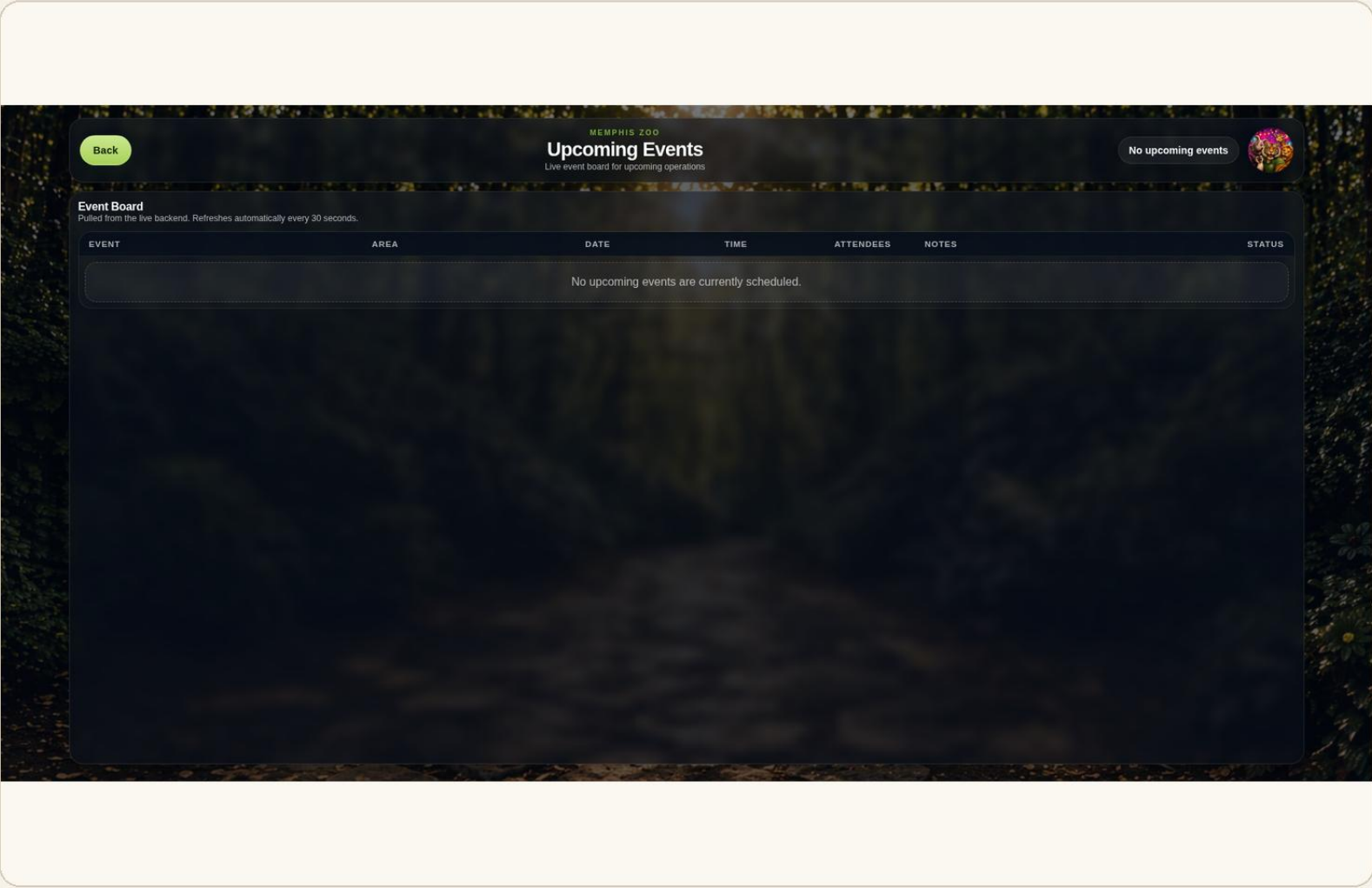
## WHAT THIS SCREEN SHOWS

- Events can be entered from a quick paste, spreadsheet import, or structured form.
- The event area matters because assigned custodians can be notified when something affects their zone.
- The public board and operational screens can pull from the same event data instead of separate lists.
- This supports “events in your area today” notifications through Memphis Messenger.



# Upcoming Events Board

PC view for the clean event board format.

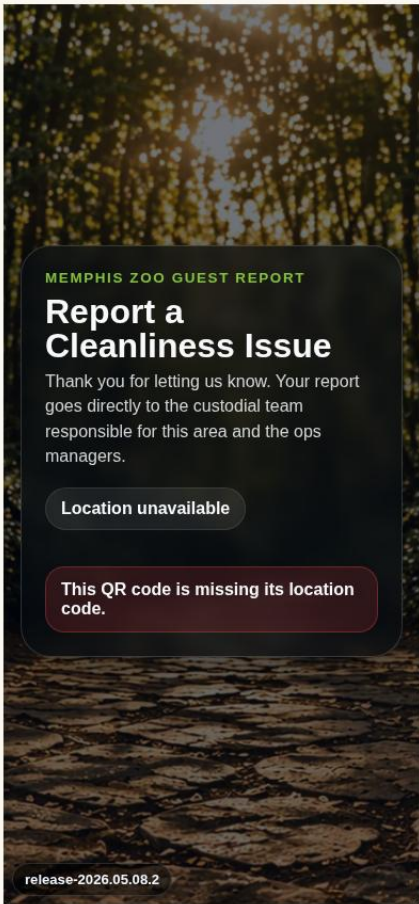


## WHAT THIS SCREEN SHOWS

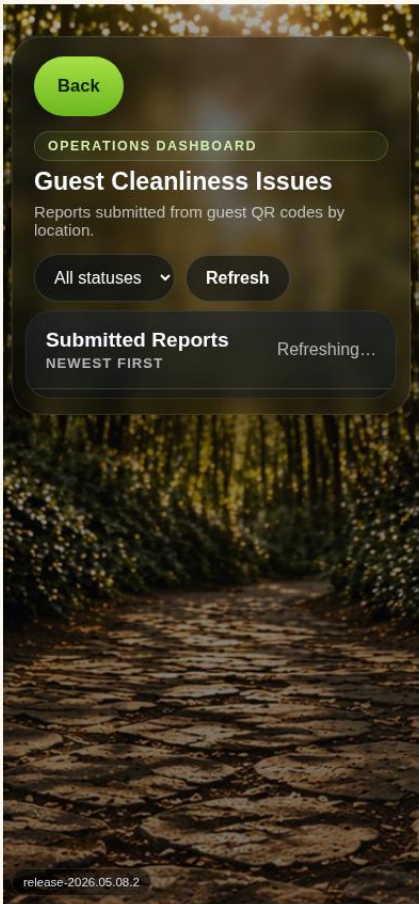
- Shows the public/manager-readable event board layout.
- Refresh behavior keeps event information aligned with backend data.
- When events exist, they can be tied back to areas so the right custodial coverage receives the right prompt.
- The empty state is useful because it makes clear when no event is currently scheduled.

# Guest Issue Reporting

A guest report should become an operations signal, not a loose comment that someone has to manually route.



Guest-facing QR form



Operations follow-up view

## NOTIFICATION PATH

- Guest scans a location QR code and submits a cleanliness issue.
- The system knows the location from the QR code and records the issue for operations visibility.
- Memphis Messenger notifies the assigned custodian and the operations contact, following the same pattern as event, due soon, and overdue notifications.
- Managers can review submitted issues from the operations screen and close the loop after follow-up.

### Why this matters

Guest-facing QR reporting turns a complaint into a routed task with a location, timestamp, and follow-up path.



# Low-Cost Setup Model

The custom path keeps the expensive part small: field hardware that is already inexpensive, plus web services that are currently running at no monthly cost.

## INITIAL SETUP ITEMS

|                        |                   |                                                                                                   |
|------------------------|-------------------|---------------------------------------------------------------------------------------------------|
| NFC tags for locations | \$45 one-time     | Scan points for restrooms, exhibits, and service locations.                                       |
| Moto G 2025 phones     | hardware only     | Phones run on zoo Wi-Fi; no carrier subscription is needed for normal operation.                  |
| Fully Kiosk licenses   | \$86 one-time     | Ten lifetime licenses to keep field phones locked into the intended workflow.                     |
| GitHub Pages           | \$0/month current | Hosts the static frontend pages.                                                                  |
| Supabase               | \$0/month current | Current backend data layer is staying on the free tier unless scale or policy needs change.       |
| Render                 | \$0/month current | Current API/backend hosting is staying on the free tier unless scale or reliability needs change. |

### Plain-English takeaway

Current monthly software cost is \$0. The realistic ongoing cost is replacing broken phones, replacing damaged/missing NFC tags, or choosing to upgrade cloud tiers later if scale, backups, reliability, or policy

## EXPANSION MODEL

- Adding more locations mainly means adding more NFC/QR points and configuration records, not buying a new software package.
- Adding more phones mainly means hardware plus kiosk licensing. Normal operation can remain Wi-Fi based.
- Backend upgrades are optional: free tiers can remain in place unless traffic, backups, reliability, or support needs justify paid service.
- The system is designed around the exact zoo workflow, so money is not spent on unused CMMS modules like parts inventory, purchasing, vendor management, or asset depreciation unless they are actually needed later.

## SOURCE NOTES

Pricing references checked: Supabase, Render, GitHub plans documentation, MaintainX, OrangeQC, UpKeep, Limble, and SafetyCulture. SaaS prices change, so final procurement should re-check vendor pages.

Most available systems cover pieces of this workflow. Matching the full combination usually requires multiple products or higher-tier plans.

**MaintainX**

**Monthly: \$20-\$75 per user on published paid tiers.**

Startup/equipment: implementation may be separate; phones/tablets, labels, and any kiosk setup are usually customer-side. Strong CMMS, but not this full zoo-specific workflow out of the box.

**OrangeQC**

**Monthly: about \$250-\$500, plus add-ons and extra inspectors.**

Startup/equipment: QR shortcuts exist, but field devices, printed codes/tags, and rollout work are still separate. Good inspection/ticket fit, not the whole custom operations hub.

**UpKeep**

**Monthly: about \$24-\$55 per user on published tiers.**

Startup/equipment: published implementation/training packages range from about \$500 to \$5,000; field devices/labels are still separate. Broad platform, more than custodial needs.

**Why the monthly cost jumps**

Commercial tools often become expensive once several paid users, QR workflows, offline mode, analytics, implementation help, request portals, API access, or multiple modules are needed.

**Initial setup difference**

Many systems do not include phones, tablets, printed QR labels, NFC tags, kiosk setup, data cleanup, or local rollout work in the monthly subscription. Those are separate startup costs.



Most available systems cover pieces of this workflow. Matching the full combination usually requires multiple products or higher-tier plans.

Limble CMMS

Monthly: quote/calculator model on public page.

Startup/equipment: QR/work request features exist, but pricing and rollout costs depend on quote; devices and labels are typically separate. Strong CMMS, broader than this use case.

SafetyCulture

Monthly: team inspection/checklist platform pricing.

Startup/equipment: mobile devices, printed QR codes, and rollout configuration are outside the base software concept. Good inspections, but not schedule + messenger + events + Memphis AI.

Is there one system this comprehensive?

Public vendor offerings cover important pieces, but one ready-made system combining zoo custodial scans, guest QR issues, events, schedule reassignment, Messenger, reminders, and Memphis AI is not obvious.

Cost conclusion

The custom path wins on cost because current monthly software cost is \$0, startup hardware is minimal, and future spending is limited to replacement phones, replacement tags, or optional cloud upgrades.

QR reporting, smart restrooms, mobile inspections, work requests, and facility dashboards are already common ideas. The difference here is the combined local workflow.

## VERIFIED PUBLIC EXAMPLE

### Georgia Tech — Smart Restroom Customer Enhancement Program

- Georgia Tech publicly describes campus-wide Smart Restrooms where users scan QR codes in participating restrooms to report issues.
- Submissions go directly to Building Services for faster response and cleaner, better-maintained restrooms.
- The program is described as nationally recognized with APPA’s Effective & Innovative Practices Award.

## BROADER MARKET EVIDENCE

- MaintainX, UpKeep, Limble, OrangeQC, and SafetyCulture all publish features around mobile work, checklists, QR/barcode scanning, reports, request portals, or inspections.
- This confirms the category is real: facilities teams are moving away from paper, radio-only workflows, and delayed inspection visibility.
- What is less common publicly is one lightweight system combining custodial area ownership, NFC work confirmation, guest QR reports, events-by-area notices, due/overdue reminders, internal messaging, and AI answers in one zoo-specific flow.

## HOW TO PHRASE THIS SAFELY

**The strongest claim is not “no one has anything like this.” The stronger and safer claim is: public products and examples prove the individual ideas are valid, but this custom system combines them around the zoo’s exact custodial operation at a much lower starting cost.**

## SOURCES CHECKED

Georgia Tech Facilities custodial page; MaintainX pricing; OrangeQC pricing; UpKeep pricing; Limble pricing/features; SafetyCulture pricing; Supabase pricing; Render pricing; GitHub plans documentation.